

Floating and sinking

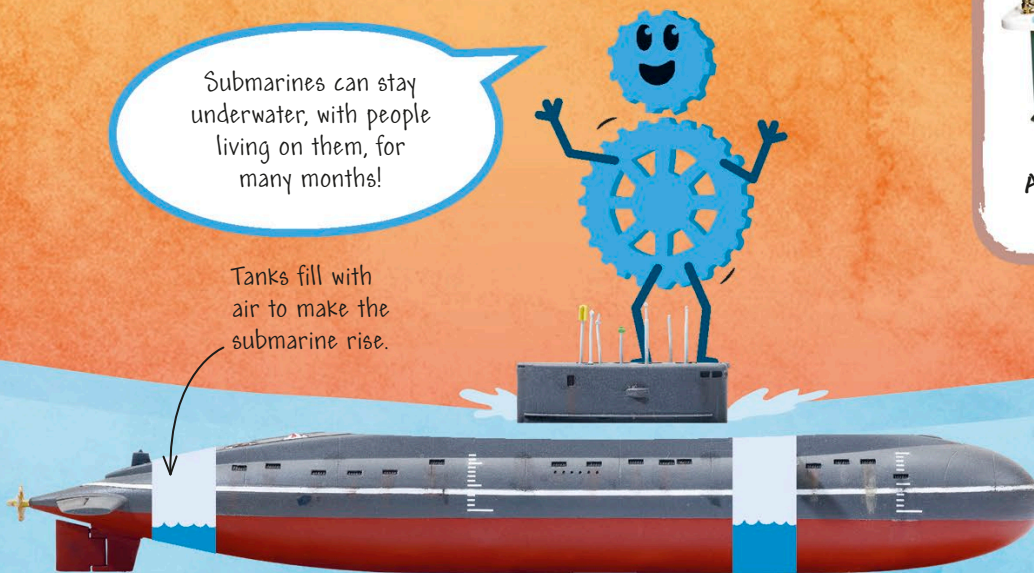
Why is a massive ship able to float when a small pebble quickly sinks? It depends on which is greater – the weight of the object in the water or the upward force of the water pushing against it.

Archimedes

Archimedes was a scientist in ancient Greece. He noticed that sinking objects push aside water. He worked out that an object will sink if it weighs more than the water it pushes aside. If it weighs less, it will float.



Archimedes figured out why things float or sink while having a bath!

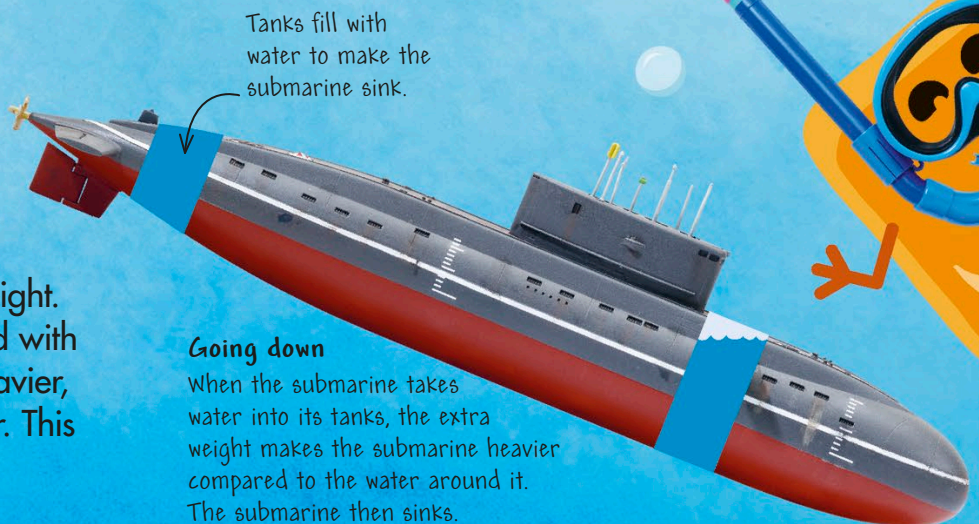


At the surface

The submarine's tanks are filled with air when it's floating at the surface.

Changing weight

Submarines can change their weight. They have tanks that can be filled with water to make the submarine heavier, or filled with air to make it lighter. This way, they can sink or rise.



Going down

When the submarine takes water into its tanks, the extra weight makes the submarine heavier compared to the water around it. The submarine then sinks.